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BRIEF REPORT

Popularity and diversity: The negative relationship in baby names in the United Kingdom

[version 1; peer review: 2 approved with reservations]

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Abstract

Background

Previous research has shown that popular names have become less popular over time. Simultaneously, accumulated evidence has indicated that names have become more diverse. However, the association between these two phenomena was unclear. This association should be revealed for a better understanding of names and naming practices. Therefore, this study investigated the relationship between the popularity and diversity of names.

Methods

I analyzed the data provided in a previous study in the U.K., which included complete records of all live births between 1996 and 2016 (N = 12,985,140).

Results

I found that the correlations between diversity and popularity indicators were highly negative, showing that they are conceptually strongly related. This means that when diversity is high, popularity is low.

Conclusions

Based on this study, we can predict one indicator from the other indicator. Because raw data on names are generally difficult to collect,

Open Peer Review

Approval Status    

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version 3				
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version 2				
(revision)	view	view	view	view
14 Jul 2025				
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11 Apr 2025	view	view		

1. Han-Wu-Shuang Bao , East China Normal University, Shanghai, China
2. Stephen J. Bush , Xi'an Jiaotong University, Xi'an, China
3. Basil Osayin Daudu ,
4. Süleyman Kasap,

Any reports and responses or comments on the article can be found at the end of the article.

this prediction is useful for understanding names and naming practices.

Keywords

popularity, diversity, name, commonality, distribution



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Introduction

Popularity of names has decreased

Previous research has indicated that popular names¹ have become less popular over time, suggesting that popularity of names has decreased (for a review, see [Ogihara, 2025](#)). For example, the rates of popular names decreased in the United States between 1880 and 2007 ([Twenge et al., 2010](#)). In this study, the rates of the top 1, 10, 25, and 50 most popular names were calculated each year, and their historical changes were analyzed (also see [Twenge et al., 2016](#)). Furthermore, [Bush et al. \(2018\)](#) demonstrated that the rates of popular names (the top 1 and 10 most popular names) decreased in the United Kingdom (England and Wales) between 1996 and 2016 (also see [Bush, 2020](#)). Similar trends were reported in Germany ([Gerhards & Hackenbroch, 2000](#)) and France ([Mignot, 2022](#)).

Not only in the West (Europe and North America) but also in the East (Asia), this trend has been observed. For instance, [Ogihara et al. \(2015\)](#) showed that the rates of popular names (the top 1, 10, 20, and 50 most popular names) decreased in Japan between 2004 and 2013 (also see [Ogihara, 2022](#)). This shift has been consistently reported ([Ogihara, 2021](#); [Ogihara & Ito, 2022](#); for a review, see [Ogihara, 2025](#)). In China ([Bao et al., 2021](#)) and Indonesia ([Kuipers & Askuri, 2017](#)), the popularity of names decreased as well.

Diversity of names has increased

At the same time, emerging evidence has shown that the diversity of names has increased over time, showing that names have become more diverse. For example, [Bush et al. \(2018\)](#) indicated that the ratio of unique (distinctive) names (the relative value of name variety) increased in the U.K. between 1838 and 2016. Moreover, this trend was observed in the U.S. between 1880 and 2017 ([He, 2020](#); also see [Mignot \(2022\)](#) for a similar report in France).

Taken together, accumulated research has indicated that names have become less popular and more diverse. [Bush et al. \(2018\)](#) has shown that these two phenomena were simultaneously observed in the U.K. between 1996 and 2016, analyzing the same dataset.

The relationship between popularity and diversity is unclear

However, the relationship between these two phenomena is unclear. Even though these two phenomena were reported within the same study ([Bush et al., 2018](#); [Mignot, 2022](#)), their relation was not directly investigated. Based on the meaning of the concepts (popularity and diversity), they are predicted to be negatively correlated. In other words, when the ratio of popular names is high, the diversity is expected to be relatively low. Similarly, when the ratio of popular names is low, the diversity is expected to be relatively high. Nevertheless, this prediction was not empirically tested. It is possible that even when the ratio of popular names is high, the diversity can also be high. For example, a population can be polarized, where some people give popular names, while others can give many varieties of unpopular names, leading to high popularity and diversity simultaneously.

This relationship between popularity and diversity should be uncovered. If the relationship is revealed, we can predict one indicator from the other. For instance, when the data and results for the top 10 most common names are available, we can infer its diversity from its popularity. In fact, this situation is frequently observed. Names are among the most private types of information. Thus, raw data on names is restricted from being openly shared, making it common for only the ranking of popular names (e.g., the top 10 most common names) to be disclosed (for a review, see [Ogihara, 2025](#)). Therefore, even when only one of the two indicators is available, we can estimate the distribution more precisely, which increases the understanding of the nature and phenomena of names and naming practices.

The current study

Therefore, in this study, I examined the relationship between the popularity and diversity of names. Specifically, I analyzed the data in the U.K. presented in previous research ([Bush et al., 2018](#)).

Based on the prior discussion above, it was predicted that there would be a negative correlation between the popularity and diversity indicators.

Method

Data

I analyzed the open data provided by [Bush et al. \(2018](#); Table S15 “Number of unique forenames, and forename diversity, in the Office for National Statistics dataset”). The data included variables on popular names and name diversity.

¹ In this article, I use the term “names” to refer to “first names” (given names, personal names, forenames), not “last names” (family names, surnames).

The original data is from the U.K. [Office for National Statistics \(2018\)](#), which included complete records of all live births in England and Wales for 21 years between 1996 and 2016. A total of 12,985,140 names were recorded, with an average of 618,340 names per year. It should be noted that names with a count of 2 or 1 were redacted to protect the confidentiality of individuals ([Office for National Statistics, 2018](#)).

Indicator

Popularity. As a popularity indicator, I used the variables “% of birth records registered with the most popular name” and “% of birth records accounted for by the top 10 names” ([Bush et al., 2018, Table S15](#)). These indicators have been used in many prior studies (e.g., [Bush, 2020](#); [Kuipers & Askuri, 2017](#); [Mignot, 2022](#); [Ogihara et al., 2015](#); [Ogihara, 2022](#); [Twenge et al., 2010, 2016](#)).

Diversity. As a diversity indicator, I used the variable “Forename diversity (i.e., ratio of the no. of unique forenames to the total no. of birth records per year)” ([Bush et al., 2018, Table S15](#)). Thus, this indicator represents the relative value of name variety. For example, when this value is 0.01, it means that there are 10 name types among 1,000 people. When this value is high, diversity is also high (there are more name types, meaning that the group is more diverse). This indicator has been used in previous research (e.g., [He, 2020](#)).

Results

Simple Pearson’s correlation coefficients among the year, popularity indicators, and diversity indicator are summarized in [Table 1](#).

Relationship within popularity indicators

The ratio of the most popular name and the ratio of the top 10 most popular names were strongly correlated, $r = .969$. This result means that these two indicators consistently measure the same concept, increasing the validity of these two indicators as name popularity indices.

Relationship between popularity and diversity

The ratio of unique names and the ratio of the most popular name between 1996 and 2016 are indicated in [Figure 1](#). As predicted, they were highly negatively correlated, $r = -.960$.

Similarly, the ratio of unique names and the ratio of the top 10 most popular names are indicated in [Figure 2](#). They were also highly negatively correlated, $r = -.994$.

Discussion

Previous research has shown that names have become less popular over time (for a review, see [Ogihara, 2025](#)). At the same time, accumulated evidence has indicated that names have become more diverse over time. However, the association between these two phenomena was unclear. This association should be revealed for a better understanding of names and naming practices. Therefore, this study investigated the relationship between the popularity and diversity of names.

I analyzed the name data provided by the previous study in the U.K. ([Bush et al., 2018](#)). I found that the correlations between the diversity and popularity indicators were highly negative, showing that they are conceptually strongly related. Specifically, in years when the ratio of unique names was high, the ratios of popular names were low. This means that when diversity is high, popularity is low. This association was very strong in the current dataset.

Based on this study, we can predict one indicator from the other indicator. We can infer diversity from popularity or popularity from diversity. Because raw data on names are generally difficult to collect, this prediction is useful for understanding names and naming practices.

Table 1. Simple Pearson’s correlation coefficients.

	Year	Popularity (top 1)	Popularity (top 10)	Diversity
Year	-	-.955	-.973	.979
Popularity (top 1)	-.955	-	.969	-.960
Popularity (top 10)	-.973	.969	-	-.994
Diversity	.979	-.960	-.994	-

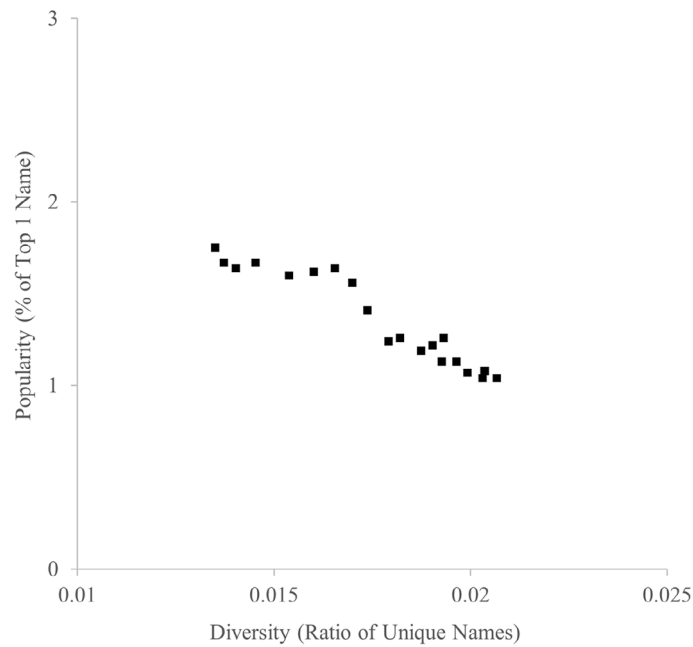


Figure 1. Diversity (ratio of unique names) and popularity (% of top 1 name) indicators of baby names in the U.K., 1996-2016.

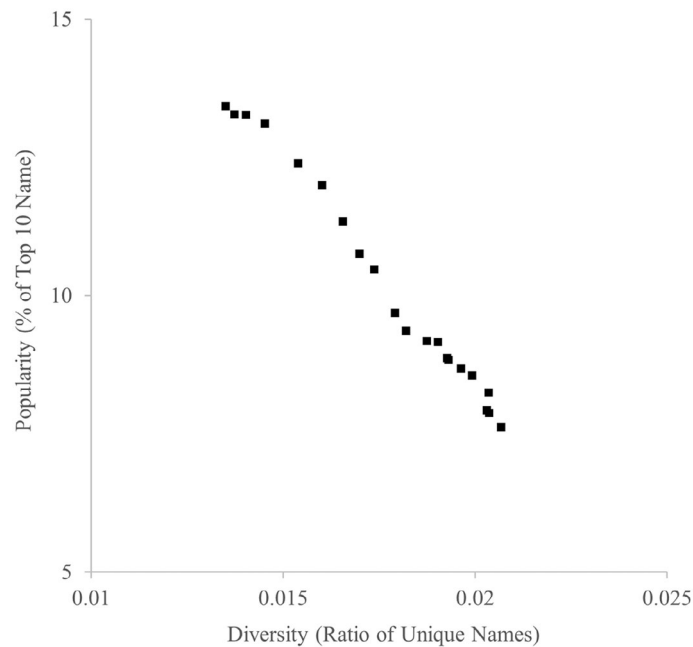


Figure 2. Diversity (ratio of unique names) and popularity (% of top 10 names) indicators of baby names in the U.K., 1996-2016.

Limitation and future direction

This study examined the relationship between diversity and popularity of names in the U.K. (England and Wales), which showed highly negative associations between them. Nevertheless, it is unclear whether this relationship is observed in other nations. Names are cultural products and are affected by many factors (e.g., [Morling, 2016](#); [Morling & Lamoreaux, 2008](#)). Thus, it is necessary to investigate this relationship in other nations.

Author contributions

The author confirms being the sole contributor of this work and approved it for publication.

Ethics and consent

Ethical approval and consent were not required.

Data availability

I analyzed the open data provided by [Bush et al. \(2018\)](#); Table S15 “Number of unique forenames, and forename diversity, in the Office for National Statistics dataset”).

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Reviewer Report 29 May 2025

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Stephen J. Bush

Xi'an Jiaotong University, Xi'an, Shaanxi, China

This is a short report which reanalyses previously published UK name data to demonstrate an inverse relationship between name popularity and name diversity, such that one variable could be used predict the other. This could provide a useful platform for future name research as many name datasets are restricted to either the top X names per year or otherwise redact rare names (about which, more below); that is, it is only practically possible to derive the first variable from them ('popularity'). In this respect, these results may facilitate the wider interpretation of smaller, and potentially overlooked, name datasets (e.g. 'top 10' lists).

Nevertheless, the result remains a single, simple, correlation analysis. Beyond the suggestions made by the first reviewer (with which I concur), the conclusion may be strengthened by some corroboratory analyses, for instance using an independent dataset sampling the same population, and/or using name datasets that do not redact rarer names (which would otherwise affect the calculations).

It is important to note that the raw data used for the analysis (Table S15 of Bush et al. 2018) ultimately derives from birth registration counts compiled by the UK Office for National Statistics (ONS); specifically,

<https://www.ons.gov.uk/file?uri=/peoplepopulationandcommunity/birthsdeathsandmarriages/livebirths/dataset> and strictly speaking covering only England and Wales rather than the entire UK. As with many name datasets, the long tail of the name distribution is redacted: in this case, names registered to < 3 people per year are excluded. As such, for any given year, the "diversity indicator" (the ratio of the no. of unique forenames to the total number of birth records per year) will be under-estimated. In addition, because the total no. of birth records will be lower than the total number of births, the "popularity indicator" (% of birth records registered with a top 1/top 10 name) will be over-estimated.

For instance, we can see from Table S15 that in 2016, the total number of birth records was 639,126 (this number being used to calculate both the popularity and diversity indicators). However, the total number of births in 2016 was 696,271, an approx. 9% difference (this value is

also made available by the ONS but in a separate file; see the link to 'download figure 1 data' in <https://www.ons.gov.uk/peoplepopulationandcommunity/birthsdeathsandmarriages/livebirths/bulletins/birthsumbirths>).

It would bolster the overall conclusion – that popularity and diversity are negatively correlated – if it could be demonstrated that refinements to these values make negligible difference. A simple way of doing so would be to calculate them using birth registration datasets that do not redact rare names in the first place. Examples present themselves in the form of data from the Government of Alberta, Canada, specifically for the years 1980-2020 (see <https://open.alberta.ca/opendata/frequency-and-ranking-of-baby-names-by-year-and-gender>; specific file: https://open.alberta.ca/dataset/11245675-b047-49fc-8bd1-cc2ce8314a6d/resource/e8aac308-c754-484c-b446-0c57ed0e8d37/download/baby-names-frequency_1980_2020.xlsx) and, to draw the focus back to the UK, data from the National Records of Scotland, covering the years 1974-2023 (<https://www.nrscotland.gov.uk/media/mbmjzs2q/full-list-1974-2023.csv>).

In addition, two points foundational to the overall argument – that “names have become less popular over time” and that “raw data on names is restricted from being openly shared, making it common for only the ranking of popular names” are supported by reference to the author’s not-yet-published review (Ogihara 2025 ‘Uncommon names are increasing globally’). To provide further context, it would help if a citation was provided for at least a pre-print of this manuscript. I am particularly drawn to the point that names have become less popular over time as, in the UK, this most strongly manifests as the 20th century progresses, i.e. during the time period covered by the ONS data. It would therefore be useful to know to what extent the relationship of popularity and diversity holds over a longer time period, and in that respect whether the conclusion can be generalised. That in mind, Bush et al. 2018 contains two different datasets sampling birth records from the English and Welsh population, of which the present work uses only one of them (from Table S15; the ONS dataset, which is much greater in depth but narrower in temporal scope). The second dataset – which has the same popularity and diversity indicators – is available in Table S6, and may be viewed as a representative population sample from 1838-2014, albeit of varying coverage depths per year. Nevertheless, a cursory examination of this data suggests that, as with the ONS data, there is a strongly negative correlation of popularity and diversity for the years 1996-2010 (this time period chosen because within it there are > 10,000 birth records per year). If we accept this as independent support of the conclusion drawn using the ONS data (and thereby the utility of this dataset), then it may be worth looking at how popularity and diversity correlate throughout history, too. This is of particular relevance as the conclusion drawn in the abstract (“we can predict one indicator from the other indicator”) implies a general relationship between the concepts that may not be universally true and that instead they may only “consistently measure the same concept” for a limited period of time.

I have expanded the Table S6 dataset substantially since the original publication; see Bush 2024 (<https://doi.org/10.5195/names.2024.2543>) and its raw data, specifically https://github.com/sjbush/uk_bmd/blob/main/dataset_B/summary_of_records_per_year.txt, which is essentially equivalent to Table S15 and contains the columns “most popular forename (% of records)” and “forename diversity (i.e., ratio of the no. of different forenames to the total no. of records per year)”. It should be clear from this data that before (approx.) the second half of the 20th century, and compared to contemporary records, forename diversity was not only persistently lower but that popular names were given to a higher proportion of people and remained popular

for longer, e.g. John and Mary were consistently the most popular male and female names, respectively, for a century (see <https://demos.flourish.studio/namehistory/?names=John,Mary>, which visualises data from Bush et al. 2018). So, although the present work (using late-20th century data) finds that one indicator predicts the other, it also appears the case that (earlier in the 20th century) popularity could vary without discernible change to diversity. Consequently, the two variables do not necessarily change simultaneously. This may be the subject of a more in-depth analysis but could be usefully commented upon here.

Finally, could some comment also be made on the gender of names, and the effect this may have on the conclusions? In contemporary UK name data, there is typically a larger pool of female than male names (discussed in Bush 2020, <https://doi.org/10.1080/00277738.2020.1775471>): that is, one would expect female names to have a higher “diversity indicator” and lower “popularity indicator” than male names. Accordingly, a direct correlation of popularity with diversity may not quite be comparing like with like: diversity being affected more strongly by female names, but popularity by male (such that in Table S15, the top 1 names, across the whole population, are all male).

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

Partly

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

Yes

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: UK naming trends and practices

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 06 Jul 2025

Yuji Ogihara

July 6th, 2025

Dear Dr. Stephen J. Bush,

Thank you very much for reviewing my manuscript and providing valuable comments.

I have modified the manuscript extensively according to the reviewers' comments.

I offer my responses to each comment below.

I have copied and pasted all of your comments without making changes.

This is a short report which reanalyses previously published UK name data to demonstrate an inverse relationship between name popularity and name diversity, such that one variable could be used predict the other. This could provide a useful platform for future name research as many name datasets are restricted to either the top X names per year or otherwise redact rare names (about which, more below); that is, it is only practically possible to derive the first variable from them ('popularity'). In this respect, these results may facilitate the wider interpretation of smaller, and potentially overlooked, name datasets (e.g. 'top 10' lists).

Thank you for your summary of this article and your words of praise.

Nevertheless, the result remains a single, simple, correlation analysis. Beyond the suggestions made by the first reviewer (with which I concur), the conclusion may be strengthened by some corroboratory analyses, for instance using an independent dataset sampling the same population, and/or using name datasets that do not redact rarer names (which would otherwise affect the calculations).

I appreciate your constructive comment. As I replied to the first reviewer, I basically agree with your opinion. Yet, at least in this article, I would like to focus on the dataset used in a previous study (details are explained below).

It is important to note that the raw data used for the analysis (Table S15 of Bush et al. 2018) ultimately derives from birth registration counts compiled by the UK Office for National Statistics (ONS); specifically, <https://www.ons.gov.uk/file?uri=/peoplepopulationandcommunity/birthsdeathsandmarriages/livebirths/d> and strictly speaking covering only England and Wales rather than the entire UK. As with many name datasets, the long tail of the name distribution is redacted: in this case, names registered to < 3 people per year are excluded. As such, for any given year, the "diversity indicator" (the ratio of the no. of unique forenames to the total number of birth records per year) will be under-estimated. In addition, because the total no. of birth records will be lower than the total number of births, the "popularity indicator" (% of birth records registered with a top 1/top 10 name) will be over-estimated.

For instance, we can see from Table S15 that in 2016, the total number of birth records was

639,126 (this number being used to calculate both the popularity and diversity indicators). However, the total number of births in 2016 was 696,271, an approx. 9% difference (this value is also made available by the ONS but in a separate file; see the link to 'download figure 1 data' in <https://www.ons.gov.uk/peoplepopulationandcommunity/birthsdeathsandmarriages/livebirths/bulletins/births>).

Thank you for your valuable note. I agree with you that this point is important.

The previous version of the article already stated that the data covered only England and Wales rather than the entire UK and that the long tail of the name distribution was redacted, as follows.

"The original data is from the U.K. Office for National Statistics (2018), which included complete records of all live births in England and Wales for 21 years between 1996 and 2016. A total of 12,985,140 names were recorded, with an average of 618,340 names per year. It should be noted that names with a count of 2 or 1 were redacted to protect the confidentiality of individuals (Office for National Statistics, 2018)." (Data section)

Thus, I have added the possible differences of the two indicators in the revised article, as follows.

"Due to the redaction of names with a count of 2 or 1, the total number of birth records (denominator) is lower than the total number of births. In contrast, the top 1 and 10 most popular names (numerator) are not influenced by the redaction. As a result, this popularity indicator is higher than the actual popularity." (Footnote 2)

"Due to the redaction of names with a count of 2 or 1, the total number of birth records (denominator) is lower than the total number of births. In contrast, the number of unique forenames (numerator) is also lower than the total number of unique forenames. Thus, this diversity indicator is different from the actual diversity, but its direction (lower or higher) is difficult to predict." (Footnote 3)

It would bolster the overall conclusion – that popularity and diversity are negatively correlated – if it could be demonstrated that refinements to these values make negligible difference. A simple way of doing so would be to calculate them using birth registration datasets that do not redact rare names in the first place. Examples present themselves in the form of data from the Government of Alberta, Canada, specifically for the years 1980-2020 (see <https://open.alberta.ca/opendata/frequency-and-ranking-of-baby-names-by-year-and-gender>; specific file: https://open.alberta.ca/dataset/11245675-b047-49fc-8bd1-cc2ce8314a6d/resource/e8aac308-c754-484c-b446-0c57ed0e8d37/download/baby-names-frequency_1980_2020.xlsx) and, to draw the focus back to the UK, data from the National Records of Scotland, covering the years 1974-2023 (<https://www.nrscotland.gov.uk/media/mbmjzs2q/full-list-1974-2023.csv>).

I appreciate your valuable comment. Datasets that do not redact rare names are especially precious because such names are frequently excluded.

Yet, this article focuses on names from England and Wales. Thus, I would like to examine these datasets in future research.

In addition, two points foundational to the overall argument – that “names have become less popular over time” and that “raw data on names is restricted from being openly shared, making it common for only the ranking of popular names” are supported by reference to the author’s not-yet-published review (Ogihara 2025 ‘Uncommon names are increasing globally’). To provide further context, it would help if a citation was provided for at least a pre-print of this manuscript. I am particularly drawn to the point that names have become less popular over time as, in the UK, this most strongly manifests as the 20th century progresses, i.e. during the time period covered by the ONS data. It would therefore be useful to know to what extent the relationship of popularity and diversity holds over a longer time period, and in that respect whether the conclusion can be generalised. That in mind, Bush et al. 2018 contains two different datasets sampling birth records from the English and Welsh population, of which the present work uses only one of them (from Table S15; the ONS dataset, which is much greater in depth but narrower in temporal scope). The second dataset – which has the same popularity and diversity indicators – is available in Table S6, and may be viewed as a representative population sample from 1838-2014, albeit of varying coverage depths per year. Nevertheless, a cursory examination of this data suggests that, as with the ONS data, there is a strongly negative correlation of popularity and diversity for the years 1996-2010 (this time period chosen because within it there are > 10,000 birth records per year). If we accept this as independent support of the conclusion drawn using the ONS data (and thereby the utility of this dataset), then it may be worth looking at how popularity and diversity correlate throughout history, too. This is of particular relevance as the conclusion drawn in the abstract (“we can predict one indicator from the other indicator”) implies a general relationship between the concepts that may not be universally true and that instead they may only “consistently measure the same concept” for a limited period of time.

I have expanded the Table S6 dataset substantially since the original publication; see Bush 2024 (<https://doi.org/10.5195/names.2024.2543>) and its raw data, specifically https://github.com/sjbush/uk_bmd/blob/main/dataset_B/summary_of_records_per_year.txt, which is essentially equivalent to Table S15 and contains the columns “most popular forename (% of records)” and “forename diversity (i.e., ratio of the no. of different forenames to the total no. of records per year)”. It should be clear from this data that before (approx.) the second half of the 20th century, and compared to contemporary records, forename diversity was not only persistently lower but that popular names were given to a higher proportion of people and remained popular for longer, e.g. John and Mary were consistently the most popular male and female names, respectively, for a century (see <https://demos.flourish.studio/namehistory/?names=John,Mary>, which visualises data from Bush et al. 2018). So, although the present work (using late-20th century data) finds that one indicator predicts the other, it also appears the case that (earlier in the 20th century) popularity could vary without discernible change to diversity. Consequently, the two variables do not necessarily change simultaneously. This may be the subject of a more in-depth analysis but could be usefully commented upon here.

Thank you for your valuable comment. Regarding the review paper, the two points (“names have become less popular over time” and that “raw data on names is restricted from being openly shared, making it common for only the ranking of popular names”) are supported not only by the review paper but also by other references I include (Bao et al., 2021; Bush, 2020; Bush et al., 2018; Gerhards & Hackenbroch, 2000; Kuipers & Askuri, 2017; Mignot, 2022; Ogihara, 2021, 2022; Ogihara & Ito, 2022; Ogihara et al., 2015; Twenge et al., 2010, 2016). Moreover, whether the review paper is made openly available would not directly affect the decision on this paper.

I also appreciate your second point. As you already acknowledged, it is unclear how the second dataset is representative (“The second dataset – which has the same popularity and diversity indicators – is available in Table S6, and may be viewed as a representative population sample from 1838-2014, albeit of varying coverage depths per year.”). In the original paper (Bush et al., 2018), the sample size for each year in the dataset is not sufficiently large and its representativeness for each year is not explained (“This approximates 130,000 to 230,000 records per year from 1838–1950, 25,000 to 100,000 records per year from 1951–2000, and 5000 to 15,000 records per year from 2001 to 2014”; p. 3). In contrast, the sample sizes are sufficiently large and representativeness of the newer dataset that I analyzed is apparently high. Thus, I analyzed the newer dataset and did not analyze the older dataset.

Finally, could some comment also be made on the gender of names, and the effect this may have on the conclusions? In contemporary UK name data, there is typically a larger pool of female than male names (discussed in Bush 2020, <https://doi.org/10.1080/00277738.2020.1775471>): that is, one would expect female names to have a higher “diversity indicator” and lower “popularity indicator” than male names. Accordingly, a direct correlation of popularity with diversity may not quite be comparing like with like: diversity being affected more strongly by female names, but popularity by male (such that in Table S15, the top 1 names, across the whole population, are all male).

I appreciate your comment. The original study (Bush et al., 2018) did not analyze name trends by gender, so this study did not either. However, I agree with your point. Thus, I have added this as a limitation, as follows.

“This study analyzed the dataset yielded by the past study (Bush et al., 2018), which did not distinguish between boys’ and girls’ names. Although a different pattern is not predicted based on gender, it is desirable to investigate the relationship between the diversity and popularity of names for boys and girls separately in the future.” (Limitation and future direction section)

Thank you for your further consideration of this manuscript.

I look forward to hearing from you at your earliest convenience.

Sincerely,

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Competing Interests: The author declares no competing interest.

Reviewer Report 06 May 2025

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Han-Wu-Shuang Bao

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In this Brief Report, the author performed a simple correlation analysis with Bush et al.'s (2018) published dataset on UK first names, showing that when the most popular names were more often used, there were also more diverse types of unique names. I think the author has addressed one of the important issues in name research and cultural change research, which lacked an empirical test before. While I believe the main argument that name popularity and name diversity can predict each other is worth publishing, I have several suggestions that may improve the strength of this argument.

First, the analysis presented in this report was simple correlation over time, which might be spurious due to covariance among year, popularity, and diversity. To more rigorously test the key relationship, I suggest either controlling for year or conducting a cross-correlation analysis and a Granger causality test. The latter two methods are usually preferred for more rigorous time series analysis in cultural change research. In doing so, the author could examine whether popularity precedes diversity, diversity precedes popularity, or they shift just concurrently. It becomes more important here because such high correlations found in this report might simply due to the fact that any linear trends over time would be highly correlated (positively or negatively) with each other.

Second, to partial out the confound by a common linear trend, I also suggest performing a simulation analysis. Specifically, you can generate or resample (e.g., bootstrap) large random samples of "names", with each sample consisting of a sufficient number of "names" simulating a sample of names in a "year". Then, for each sample, you can calculate the indices of popularity and diversity. Finally, across these simulated samples, you can test the correlation between popularity and diversity. In this way, you can test this relationship without the time confound. This may also

answer if there is a purely statistical relationship between popularity and diversity, providing a better understanding of the main research question here.

Besides these suggestions, I have one minor comment on including more past evidence for name diversity. Bao et al. (2021) indeed tested changes in six name indices in China, where “standard deviation of name length” and “proportion of three-character given names” can both be regarded as indices of name diversity in Chinese naming practices. So I suggest adding them as another piece of evidence into the section “Diversity of names has increased”.

Is the work clearly and accurately presented and does it cite the current literature?

Yes

Is the study design appropriate and is the work technically sound?

No

Are sufficient details of methods and analysis provided to allow replication by others?

Yes

If applicable, is the statistical analysis and its interpretation appropriate?

No

Are all the source data underlying the results available to ensure full reproducibility?

Yes

Are the conclusions drawn adequately supported by the results?

Partly

Competing Interests: No competing interests were disclosed.

Reviewer Expertise: cultural change, names, naming practices, name uniqueness

I confirm that I have read this submission and believe that I have an appropriate level of expertise to confirm that it is of an acceptable scientific standard, however I have significant reservations, as outlined above.

Author Response 06 Jul 2025

Yuji Ogihara

July 6th, 2025

Dear Dr. Han-Wu-Shuang Bao,

Thank you very much for reviewing my manuscript and providing valuable comments.

I have modified the manuscript extensively according to the reviewers' comments.

I offer my responses to each comment below.

I have copied and pasted all of your comments without making changes.

In this Brief Report, the author performed a simple correlation analysis with Bush et al.'s (2018) published dataset on UK first names, showing that when the most popular names were more often used, there were also more diverse types of unique names. I think the author has addressed one of the important issues in name research and cultural change research, which lacked an empirical test before. While I believe the main argument that name popularity and name diversity can predict each other is worth publishing, I have several suggestions that may improve the strength of this argument.

Thank you for your words of praise. Following your comments below, I have modified the manuscript.

First, the analysis presented in this report was simple correlation over time, which might be spurious due to covariance among year, popularity, and diversity. To more rigorously test the key relationship, I suggest either controlling for year or conducting a cross-correlation analysis and a Granger causality test. The latter two methods are usually preferred for more rigorous time series analysis in cultural change research. In doing so, the author could examine whether popularity precedes diversity, diversity precedes popularity, or they shift just concurrently. It becomes more important here because such high correlations found in this report might simply due to the fact that any linear trends over time would be highly correlated (positively or negatively) with each other.

Thank you for your valuable comment. I agree with you that it is important to consider this point carefully.

I do not claim that there is a causality between the two indicators. As I write in the article, it is important to predict one indicator from the other indicator in name research. It is not necessary to determine whether popularity precedes diversity, diversity precedes popularity, or they shift just concurrently. Thus, I think simple correlations are sufficient, at least in this context. Nevertheless, I also think that the analyses you suggested work well in cultural change research.

However, if you do not find my response to be sufficient, I will reconsider this point. I would appreciate it very much.

Second, to partial out the confound by a common linear trend, I also suggest performing a simulation analysis. Specifically, you can generate or resample (e.g., bootstrap) large random samples of "names", with each sample consisting of a sufficient number of "names" simulating a sample of names in a "year". Then, for each sample, you can calculate the indices of popularity and diversity. Finally, across these simulated samples, you can test the correlation between popularity and diversity. In this way, you can test this relationship without the time confound. This may also answer if there is a purely statistical relationship between popularity and diversity, providing a better understanding of the

main research question here.

I appreciate your constructive comment. That is a fascinating idea. I agree that such a simulation analysis can contribute to a better understanding of names and naming practices.

In contrast, I think that the approach is beyond the scope of this article which analyzes the actual data from the United Kingdom.

Moreover, this simulation could tend to lack ecological validity. The results should strongly depend on parameters of a hypothetical and imaginary dataset.

Furthermore, this simulation lacks a time perspective. Our focus is on how names (and name indicators) change over time.

Therefore, I would like to consider this new idea in a different study. Anyway, thank you very much for your important input.

Besides these suggestions, I have one minor comment on including more past evidence for name diversity. Bao et al. (2021) indeed tested changes in six name indices in China, where "standard deviation of name length" and "proportion of three-character given names" can both be regarded as indices of name diversity in Chinese naming practices. So I suggest adding them as another piece of evidence into the section "Diversity of names has increased".

Thank you for your comment. I agree that the indicators you mention are related to name diversity. Yet, these indicators do not directly show name diversity because they focus on "name length." Even when the standard deviation of name length increases, name diversity can decrease. For example, even when names consisting of one or three characters increase, if the same names are given, diversity can decrease. Therefore, I maintain this part.

Thank you for your further consideration of this manuscript.

I look forward to hearing from you at your earliest convenience.

Sincerely,

Yuji Ogihara, Ph.D.
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Competing Interests: The author declares no competing interest.

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